

RCIM Model-Bank Reproduction Forward Final Open-Cells Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

- campaign name: track1_forward_final_open_cells_campaign_2026-04-28_00_30_09
- planning report: doc/reports/campaign_plans/track_1/exact_paper/2026-04-28-00-16-01_track1_forward_final_open_cells_campaign_plan_report.md
- queue size: 76
- execution window: 2026-04-28T00:32:27.4229313+02:00 to 2026-04-28T02:20:01+02:00
- campaign output directory: output/training_campaigns/track1/exact_paper/forward_final_open_cells/track1_forward_final_open_cells_campaign_2026-04-28_00_30_09
- validation root: output/validation_checks/paper_reimplementation_rcim_original_dataset_exact_model_bank_forward_final_open_cells
- report timestamp: 2026-04-28-11-00-53

Executive Summary

- The final forward-only residual wave completed the full 76/76 queue successfully.
- Targeted family-target pairs: 8 .
- Promoted pair winners: 2 .
- Retained baseline pair winners: 6 .
- Forward non-green cells moved from 11 to 10 .

Family Best Retry Outcome

Family	BestRun	Scope	Harmonic	ClosureScore	Met	Near	Open
ERT	track1_forward_ert_ampl_h162_final_open_cells_attempt_08	amplitude	162	0.500	1	0	1
GBM	track1_forward_gbm_ampl_h162_final_open_cells_attempt_08	amplitude	162	0.000	0	0	2
XGBM	track1_forward_xgbm_ampl_h240_final_open_cells_attempt_08	amplitude	240	0.750	1	1	0
LGBM	track1_forward_lgbm_phase_h81_final_open_cells_attempt_05	phase	81	1.000	2	0	0

Campaign Best Run

- run: track1_forward_lgbm_phase_h81_final_open_cells_attempt_05
- family: LGBM
- scope: phase
- harmonic: 81
- closure score: 1.000
- met / near / open: 2 / 0 / 0

Forward Benchmark Delta

Surface	Before	After	Delta
Table 2 forward amplitude MAE	94G / 3Y / 3R	93G / 4Y / 3R	1Y / 0R
Table 3 forward amplitude RMSE	96G / 3Y / 1R	97G / 2Y / 1R	-1Y / 0R
Table 4 forward phase MAE	89G / 1Y / 0R	90G / 0Y / 0R	-1Y / 0R
Table 5 forward phase RMSE	90G / 0Y / 0R	90G / 0Y / 0R	0Y / 0R

Canonical Forward Status After Closeout

- Table 2 : 4 yellow, 3 red
- Table 3 : 2 yellow, 1 red
- Table 4 : 0 yellow, 0 red
- Table 5 : 0 yellow, 0 red

Reference Archive Refresh

Direction	Family	Archived Targets	Unique Source Runs	Unique Source Configs	Archive Root
forward	SVM	19	11	11	models/paper_reference/rcim_track1/forward/svm_reference_models
forward	MLP	19	13	13	models/paper_reference/rcim_track1/forward/mlp_reference_models
forward	RF	19	12	12	models/paper_reference/rcim_track1/forward/rf_reference_models
forward	DT	19	12	12	models/paper_reference/rcim_track1/forward/dt_reference_models
forward	ET	19	12	12	models/paper_reference/rcim_track1/forward/et_reference_models
forward	ERT	19	10	10	models/paper_reference/rcim_track1/forward/ert_reference_models
forward	GBM	19	11	11	models/paper_reference/rcim_track1/forward/gbm_reference_models
forward	HGBM	19	11	11	models/paper_reference/rcim_track1/forward/hgbm_reference_models
forward	XGBM	19	12	12	models/paper_reference/rcim_track1/forward/xgbm_reference_models
forward	LGBM	19	17	17	models/paper_reference/rcim_track1/forward/lgbm_reference_models

Linked Artifacts

- benchmark report: `doc/reports/analysis/rcim_paper_reference/RCIM Paper Reference Benchmark.md`
- training results master summary: `doc/reports/analysis/Training Results Master Summary.md`
- track1 reference root: `models/paper_reference/rcim_track1/forward/`
- campaign leaderboard:
`output/training_campaigns/track1/exact_paper/forward_final_open_cells/track1_forward_final_open_cells_campaign_2026-04-28_00_30_09/campaign_leaderboard.yaml`
- campaign best run YAML:
`output/training_campaigns/track1/exact_paper/forward_final_open_cells/track1_forward_final_open_cells_campaign_2026-04-28_00_30_09/campaign_best_run.yaml`
- campaign best run Markdown:
`output/training_campaigns/track1/exact_paper/forward_final_open_cells/track1_forward_final_open_cells_campaign_2026-04-28_00_30_09/campaign_best_run.md`